

Outer Race Bearing Replacement — Gearbox

Stage 3 Outer Race Fault — Bottleneck Asset with Zone D Isolation

Asset: AST_GBX_001 (Gearbox Unit) | LINE_001 | SKF22318

Bearing Type: SKF22318 | Est. Duration: 8 h | Crew: 2 | Priority: CRITICAL

SOP_005 | v1.0 | Revised: 2026-04-10

WARNING — CRITICAL ASSET: AST_GBX_001 is a high-criticality bottleneck on LINE_001 with downtime cost of INR 18,000/hour and Zone D ISO 10816 classification. No personnel may approach this asset without LOTO_004 fully applied. Financial exposure at Stage 3: approximately INR 122,400 if run to failure.

1. Purpose and Scope

This procedure covers planned replacement of the input or output shaft bearing on AST_GBX_001 (Flender FL-GBX-88) where an outer race fault at Stage 3 has been confirmed. Stage 3 is defined as BPFO energy exceeding 3.5× baseline AND kurtosis exceeding 5.0. The part required is PART_003 (SKF22318-E). CRITICAL: PART_003 has a standard lead time of 30 days and an expedited lead time of 12 days. If stock is zero, DO NOT open the gearbox before the replacement bearing is on site. Opening the gearbox without the replacement part available will cause a forced extended outage.

2. Prerequisites

- CRITICAL: PART_003 (SKF22318-E) physically on site and confirmed. Do NOT start disassembly without the replacement bearing in hand.
- If PART_003 qty_on_hand = 0: initiate emergency procurement. Estimated expedited lead: 12 days. Do not open gearbox until part arrives.
- LOTO_004 confirmed current (is_current = true). Two technicians available, both LOTO_CERTIFIED.
- At least one technician holding ISO_CAT_III for Zone D access and post-repair vibration validation.
- Approved maintenance window of minimum 8 hours (CTX_001 or equivalent).
- Zone D access confirmed — all personnel on shift briefed on Zone D isolation requirement.
- Oil drain and collection system confirmed available.
- Induction heater, bearing puller, and calibrated torque wrench on site.
- Operating load reduced by 15–20% during monitoring period before repair (derate procedure).

3. Safety Requirements

3.1 LOTO Isolation Procedure

LOTO Permit	LOTO_004
Isolation Points	Gear-Isolation-1, Oil-Drain-GBX
Energy Types	Mechanical, Oil System
Zone Classification	D — No approach without LOTO fully applied. Non-negotiable.
Isolation Steps	Apply mechanical isolation at Gear-Isolation-1. Open oil drain at Oil-Drain-GBX to release hydraulic pressure. Lock and tag both points. Each technician applies personal padlock.
Verification	Attempt to operate input shaft by hand — must not move. Confirm zero oil pressure at gauge. Confirm all drive power isolated.

3.2 PPE Requirements

Mandatory PPE	Full face shield, heavy-duty nitrile gloves, oil-resistant apron, steel-toed safety boots, safety helmet
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Oil Spill Risk	Zone D oil system — have absorbent pads and drip trays positioned before any oil system components are opened
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4. Tools and Materials Required

Part Number	Description	Quantity	Bin Location
PART_003	SKF22318-E Spherical Roller Bearing	1	BIN_C1 (CHECK STOCK — may be 0)
PART_005	Industrial Oil ISO68 (for re-fill after service)	500 mL	BIN_L2

- Heavy-duty bearing puller (rated for SKF22318 mass)
- Induction heater rated for large-bore bearings (target 100°C for SKF22318)
- Torque wrench, calibrated, range 50–400 Nm
- Dial indicator and magnetic base (for shaft runout check)
- Oil drain container (minimum 2 L for gearbox sump)
- Oil sample vial for pre-drain oil condition record
- Portable vibration analyser
- Infrared thermometer
- Clean lint-free rags and industrial solvent
- Precision feeler gauges for bearing clearance check

5. Step-by-Step Procedure

1	CRITICAL STOP: Confirm PART_003 is physically in hand. If not: STOP. Do not proceed. Initiate emergency procurement and monitor under derate.
2	Confirm maintenance window is active and duration ≥ 8 hours. This job requires uninterrupted 8-hour block — do not start if window is less than 8 hours.
3	Apply LOTO_004 per Section 3.1. Each technician applies personal padlock. Verify zero energy and zero oil pressure.
4	Drain gearbox oil via Oil-Drain-GBX. Collect sample for condition record (colour, particles, smell).
5	Remove gearbox access cover (side or top as per FL-GBX-88 manual section 4.2).
6	Remove input shaft bearing retaining ring or end plate as applicable.
7	Using heavy-duty bearing puller, extract existing SKF22318-E bearing. Note any abnormal resistance.
8	Inspect removed bearing: photograph spalling, measure raceway damage, note oil condition inside bearing.
9	Inspect shaft journal and housing bore with feeler gauges. Check shaft runout with dial indicator. Record all measurements.
10	Clean housing bore and shaft journal thoroughly with lint-free cloth and solvent.
11	Heat new SKF22318-E in induction heater to 100°C. Do not exceed 120°C.
12	Slide heated bearing onto shaft smoothly — press force on inner ring only. Allow to cool.
13	Refit retaining ring or end plate. Torque to specification (see Section 6).
14	Refit gearbox access cover. Torque cover bolts to specification.
15	Refill gearbox oil with fresh PART_005 to fill-line mark (approximately 400–500 mL depending on housing).
16	Remove LOTO_004. Controlled restart at reduced load (50%) for 1 hour.
17	Monitor vibration and temperature at 30 minutes. Record on work order.
18	Ramp to full load. Take measurements at 1 hour.
19	Final QA check at 24 hours: confirm all criteria in Section 7 are met.

20	Update bearing_master. Record actual duration and cost. Close work order.
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6. Critical Specifications

Retaining Ring / End Plate Torque	180 Nm — torque wrench required, not impact gun
Cover Bolt Torque	95 Nm in cross pattern
Bearing Fit	Interference fit — induction heating required
Mounting Temperature	100°C target, max 120°C
Oil Type	PART_005 — Industrial Oil ISO68
Oil Fill Quantity	400–500 mL — fill to mark, not by volume alone
Shaft Runout Acceptance	≤ 0.02 mm TIR — if exceeded, do not fit new bearing until shaft is inspected

7. Post-Repair Quality Acceptance Criteria

Parameter	Acceptance Criterion	Failure Action
Vibration RMS (at full load)	≤ 3.5 mm/s	Shut down, re-inspect bearing seating
Peak Vibration	≤ 4.0 g	Shut down, re-inspect
Bearing Temperature	≤ 65°C at 1 h ≤ 62°C at 24 h	Check oil level; re-inspect if rising
Kurtosis at 24 h	≤ 2.8	Re-inspect bearing seating and shaft alignment
BPFO Energy (normalised)	≤ 1.2× baseline	Re-inspect — bearing may be misseated
Oil Condition after 24 h	No new metallic particles in sump sample	Shut down, full teardown

8. Part Shortage Protocol (PART_003 qty = 0)

If PART_003 stock is zero at the time of fault detection, the following interim protocol applies until the replacement bearing arrives (expedited lead time: 12 days):

- Initiate emergency procurement of PART_003 from SKF India — expedited delivery (12 days).
- Reduce gearbox operating load by 15–20% to extend RUL under reduced stress.
- Increase monitoring frequency to every 2 hours (historian polling interval).
- Pre-book maintenance window for Day 13 after today (1 day after part arrives).
- Brief all shift supervisors: Zone D — no approach without LOTO_004 at any time.
- If vibration exceeds 10 mm/s or kurtosis exceeds 9.0 before part arrives: initiate controlled shutdown.
- Do NOT open the gearbox without the replacement bearing physically on site.

9. Related Documents

- SOP_006 — Post-Repair QA Procedure (all assets)
- LOTO_004 permit record — AST_GBX_001 (Zone D)
- CASE_004 — Stage 3 outer race, AST_GBX_001, part shortage, monitoring active
- CASE_012 — Lubrication service, AST_GBX_001, BRG_012 (OUTPUT), oil changed preventively
- Flender FL-GBX-88 maintenance manual — Section 4 (bearing replacement)